Self-Fulfilling Expectations in Evolutionary Games

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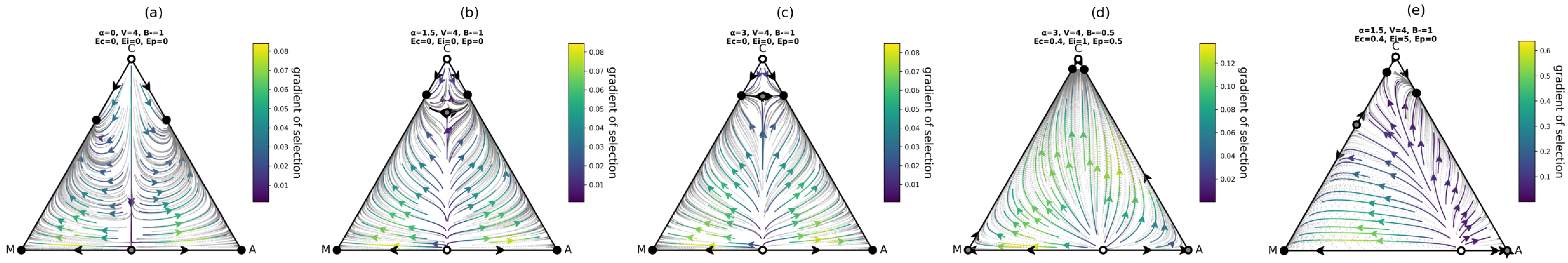

Figure 1. Each ternary plot maps the evolutionary trajectory of a three-strategy population, where the background heatmap tracks the gradient of selection from weak (dark purple) to strong (bright yellow). Stationary points are color-coded by type: stable sinks (black), unstable sources (white), and saddle points (grey).

Summary

This study adopts an Evolutionary Game Theory (EGT) framework to investigate the conditions under which social conventions remain self-reinforcing or collapse under adversarial pressure, in which the population's own composition endogenously determines the information environment that agents use to form expectations and select strategies. While our previous agent-based simulations captured the emergent micro-level dynamics of herd-like lock-in arising from distributed and subjective expectations, here we examine macro-level stability and structural collapse as a complementary approach [1]. By extending EGT with an endogenous social field (P), we examine how enforcement, subversion, and conformity interact under dynamically generated perceptions of "structural clarity."

Introduction

Self-organizing multi-agent systems coordinate through socially constructed arrangements that evolve over time [2]. A central example is the *convention*: a pattern of behavior grounded in mutually held expectations [3]; emerging as a regularity of outcomes in recurring coordination problems [4]. These conventions are self-reinforcing and shape collective outcomes without requiring central coordination [5]. However, long-term systemic stability is fragile, dependent upon the population composition and the legibility of the social environment.

To understand how decentralized regimes persist, we draw on Bicchieri's account of social conventions [3]. Conventions are stable patterns of behavior sustained by shared empirical expectations [3, p.34]; though they need not be symmetric.

In many settings, social order relies on a class of "civic" agents who actively uphold institutional order by engaging in social maintenance. These actors monitor behavior, respond to violations, and reinforce collective expectations, despite incentives to free-ride. As such, their role has been widely examined in the literature on governance, cooperation, and institutional maintenance [1, 6].

Whereas civic agents sustain order through active monitoring and enforcement, conformers sustain order through adaptive alignment with prevailing behavior. In Bicchieri's framework [3], empirical expectations sustain conventions because individuals comply in part because they expect others to do so.

While conventions are inherently resistant to change, transitions can occur rapidly through cascades and tipping points [7]. Because conventions are self-enforcing, deviation is individually costly. Change therefore depends on early movers who challenge prevailing expectations and make alternatives visible, shifting collective beliefs [3].

Model Definitions

We present an evolutionary model comprising Maintainers (M), Conformers (C), and Adversaries (A), in which perceived social structures and behavioral choices co-evolve.

A central model object is the social field $P \in [0, 1]$, representing perceived dominance of M or A. High values correspond to stable, maintainer-dominated regimes, low values correspond to adversarial dominance, and intermediate values correspond to socially ambiguous or contested regimes. For each discrete group configuration sampled from the simplex, we define the local strategy fractions as $x_M = N_M/G$, $x_C = N_C/G$, and $x_A = N_A/G$. The initial field perception $P^{(0)}$ generated within the group is given by:

$$P^{(0)} = \sigma\left(k(x_M - x_A)\right), \tag{1}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ represents the standard logistic function, and k controls the sensitivity of the field to differences between maintainers and adversaries.

Table 1. Parameters for the Endogenous Social Field Model

Sym.	Role	Domain/Values
k	Perceptual Sensitivity	$\{10\}$
α	Conformer Plasticity	$\{0, 1.5, 3\}$
B	Maintainer Benefit	$\{1.0\}$
E_c	Enforcement Cost	$\{0, 0.2, 0.4, 0.6, 0.8, 1\}$
E_i	Enforcement Effectiveness	$\{0, 1, 3, 5\}$
B^-	Adversary Benefit	$\{0.5, 1, 1.5\}$
E_p	Enforcement Leakage	$\{0, 0.2, 0.5\}$
V	Coordination Weight	$\{1, 2, 4\}$

Strategies

Maintainers (M): Ideologues committed to preserving a focal convention through active enforcement. They pay a fixed enforcement cost (E_c) and receive an intrinsic benefit B proportional to the effective compliant population ($\tilde{x}_M$). Enforcement effectiveness (E_i) determines suppression of adversaries, while enforcement cost (E_c) is modeled separately to capture inefficiencies between expenditure and enforcement outcomes.

Adversaries (A): Ideologues seeking to replace the existing convention with an alternative equilibrium. They receive intrinsic benefit B^- proportional to the effective support for the alternative ($\tilde{x}_A$). Maintainer suppression is

$$\Psi = x_M E_i, \tag{2}$$

assuming enforcement capacity scales with the prevalence of M. Adversaries also gain from enforcement leakage,

$$L = E_p x_M E_c, \tag{3}$$

where E_p is the leakage coefficient.

Conformers (C): Socially reactive agents with no intrinsic ideological preference. They estimate the relative prevalence of M and A and adopt A with probability x_{CA} (otherwise M), where switching is governed by plasticity α. Their utility depends only on coordination,

$$V(P^{(1)} - 0.5)^2, \tag{4}$$

where $V \in \{1, 2, 4\}$ scales coordination benefits. Conformers neither receive ideological rewards (B, B^-) nor pay enforcement costs, but those aligning with A incur expected suppression $x_{CA}\Psi$.

Configuration Payoff Formulations

The local configuration payoff values for each strategy within a specific group configuration are mapped directly into the `MatrixNPlayerGameHolder` matrix. These local payoff functions (f_M, f_C, f_A) are defined as:

$$\begin{aligned} f_M &= (B \cdot \tilde{x}_M) - E_c, \\ f_C &= V \cdot (P^{(1)} - 0.5)^2 - (x_{CA} \cdot \Psi), \\ f_A &= (B^- \cdot \tilde{x}_A) + L - \Psi. \end{aligned} \tag{5}$$

Simulation

We simulated all 1,944 parameter combinations, each from 100 initial population states. Because state-dependent payoffs generate high-order polynomial fitness landscapes, standard solvers can slow near transitions and terminate prematurely at boundary layers. We therefore used a two-step verification protocol: a 5% spatial threshold clustered nearby roots into the same macro-attractor basin (`atol_equal` in `egttools`), and boundary equilibria were manually verified by local perturbation to correct for numerical instability near corners and edges.

Table 2. Summary of stable points and their pairwise overlap across all simulations. Percentages do not sum to 100%, a single parameter combination can exhibit multiple stable basins

Stable Point	Count	Percentage in Outcomes
A (Vertex-Pure)	1520	78.2%
M (Vertex-Pure)	1455	74.8%
A–C (Edge-Mixed)	517	26.6%
M–C (Edge-Mixed)	118	6.1%

Comparison	Both	First Only	Second Only	Neither
A (VP) vs. M (VP)	1177	343	278	146
A (VP) vs. A–C (EM)	199	1321	318	106
M (VP) vs. M–C (EM)	103	1352	15	474

Table 2 summarizes the prevalence of stable equilibria across simulations. Vertex-pure equilibria dominated, with stable A and M occurring in 78.2% and 74.8% of parameter combinations, respectively, whereas edge-mixed equilibria were less common, particularly along the M–C edge (6.1%). The edge-mixed equilibria ($A-C$ and $M-C$) capture states of dynamic coexistence where Conformers explicitly balance the social field alongside a dominant strategy. Our results suggest that in systems governed by state-dependent feedback, conformity amplifies polarization between A and M instead of stabilizing intermediate states.

Leaderless Sheep and Pure Conformity

Pure conformity is inherently unstable. Without dedicated Maintainers or Adversaries to anchor the social field, conformers face maximal ambiguity, making the population highly susceptible to committed change (Figure 1).

Polarization, Bifurcation and Multi-stability

Figure 1 shows how conformer plasticity (α) reshapes the evolutionary landscape. With no plasticity (a), populations collapse toward pure conventions along a central repelling manifold. Introducing moderate (b) and high (c) behavioral plasticity fundamentally reorganizes the phase space, generating internal saddle bifurcations and creating stable, conformer-permissive boundary sinks. As plasticity scales, it replaces the need for large, costly blocks of dedicated ideologues, allowing minute behavioral anchors to successfully lock in population-wide coordination.

Panels (c) and (d) collectively illustrate a profound trade-off between local equilibrium stability and global structural resilience. Under high plasticity ($\alpha = 3$), stable sinks compress tightly against the pure conformer ceiling, leaving the population balancing on a razor-thin topological edge. Panel (d) shows that penalizing pure strategies triggers a tipping point toward a conformer-permissive convention anchored by a minimal committed minority. Though stable conventions remain closely spaced indicating limited resilience.

Finally, panel (e) demonstrates that institutional asymmetries distort the phase space by shifting edge equilibria: maintenance costs raise the coordination threshold along the M–C edge, while suppression penalties push the C–A equilibrium toward a more committed adversarial state.

Parameter Effects on Emergent Regimes

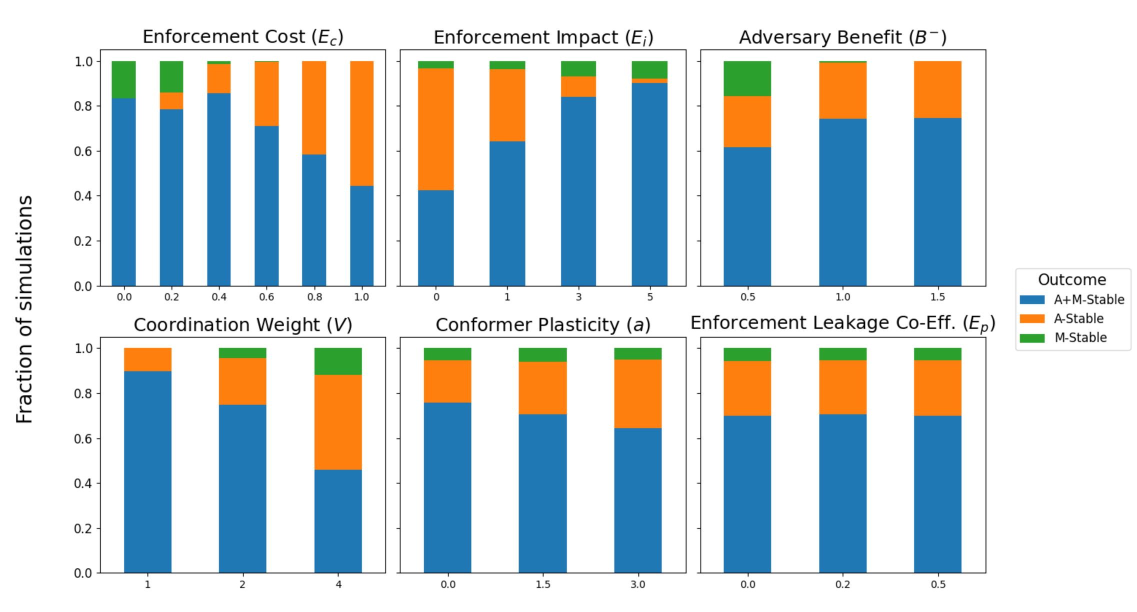

Figure 2. Parameter dependence of stability regimes. A and M denote exclusive attractors; "$A + M$" indicates bistability.

We examined how model parameters (Table 1) shape emergent stability regimes. Because outcomes arise from parameter interactions, Figure 2 shows marginal trends rather than independent causal effects.

Institutional parameters govern macro-stability. Higher enforcement cost (E_c) and coordination weight (V) reduce multi-stability in favor of exclusive regimes, whereas greater enforcement effectiveness (E_i) expands multi-stability. The combination of high enforcement cost and zero effectiveness ($E_c = 0.8$–1.0, $E_i = 0$) uniquely eliminates multi-stability, producing exclusively A-Stable outcomes. **Behavioral parameters primarily reshape coordination pathways.** Increasing conformer plasticity (α) gradually shifts outcomes from A+M-Stable toward A-Stable while acting as a topological catalyst for coordination dynamics. By contrast, the leakage coefficient (E_p) has little effect on macro-stability because leakage operates only during transient Maintainer–Adversary coexistence effecting only trajectories.

References

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